Module 1: Overview & Transmission Fundamentals

OSI vs. TCP/IP, Impairments, Nyquist & Shannon Theorems

Module I: Data Communications & Networking Overview — Topics Covered

1. Five Components of Data Communication
2. Network Topologies (Mesh, Star, Bus, Ring)
3. OSI 7-Layer vs. TCP/IP 4-Layer Architecture
4. Transmission Impairments (Attenuation, Delay, Noise)
5. Nyquist Maximum Data Rate Theorem
6. Shannon Channel Capacity Theorem & SNR (dB)

1. Layered Network Models: OSI vs. TCP/IP

OSI Layer	Protocol Data Unit (PDU)	Key Responsibilities	TCP/IP Equivalent
7. Application	Messages / Data	Network virtual terminal, HTTP, FTP, SMTP, DNS	**Application Layer**
6. Presentation	Formatted Data	Translation, Encryption (SSL/TLS), Compression	
5. Session	Dialog Tokens	Session establishment, Dialog control, Sync checkpoints	
4. Transport	Segments (TCP) / Datagrams (UDP)	Port addressing, Segmentation, Flow/Error/Congestion control	**Transport Layer**
3. Network	Packets	Logical IP addressing, Subnet routing, Path determination	**Internet Layer**
2. Data Link	Frames	Physical MAC addressing, Framing, Error detection (CRC)	**Network Access Layer**
1. Physical	Bits	Bit transmission over copper/fiber/radio, signal levels	

2. Channel Capacity Theorems

2.1 Nyquist Theorem (Noiseless Channel)

$$C = 2B\log_2(M) \text{ bps}$$

where B is channel bandwidth in Hertz and M is the number of discrete signal voltage levels.

2.2 Shannon Theorem (Noisy Channel)

$$C = B\log_2(1 + \text{SNR}) \text{ bps}$$

where $\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$. Given $\text{SNR}_{\text{dB}} = 10\log_{10}(\text{SNR}) \implies \text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)}$.

🏛️ **BIT Mesra Mid-Sem Exam Numerical (8 Marks)**

Problem: A telephone line has a bandwidth of $4\ \mathrm{kHz}$ and a signal-to-noise ratio of $30\ \mathrm{dB}$. Calculate the theoretical Shannon maximum channel capacity.

Solution:

$$\mathrm{SNR}_{\mathrm{dB}} = 30 \implies 10\log_{10}(\mathrm{SNR}) = 30 \implies \mathrm{SNR} = 10^3 = 1000$$

$$C = B\log_2(1+\mathrm{SNR}) = 4000 \times \log_2(1001) \approx 4000 \times 9.967 = \mathbf{39{,}869}\ \mathrm{bps} \approx \mathbf{39.87}\ \mathrm{kbps}$$